Project assignment

The aim of the project is the exploration, training and serving of a machine learning model for either Classification or Time Series Data forecasting to be delivered as SAAS on a secure cloud infrastructure.

There are three options for your project:

1. Forecasting the insuline intake for patients affected by Type 1 diabetes
2. Discriminating healthy subject from Parkinsons's disease patients
3. Discriminating healthy subject from Alzheimer's disease patients

Project goals

There are five primary goals for your project:

- Demonstrate an End-to-End containerized application example on the Cloud
- Present the application and the ML model chosen
- Build a web app able to perform the task, supposing that the APP should serve a total number of about 50,000 customers and, during the serving time, with up to 5,000 simultaneous service requests per minute.
- Discuss and motivate how to set the cluster security measures depending on the chosen specific application and provide their implementation by using NGINX.

Project structure

To this end, you should perform the following tasks:

A) Train an ML model offline;

B) Save the final model, and build a WEB APP able to perform the forecasting on new data;

C) Containerize the APP and serve it on Docker Desktop;

D) Check the vulnerabilities using Docker Scout;

E) Eventually, patch the WEB APP, detaching all the vulnerabilities that it is possible to remove

F) Create a Kubernetes cluster with one control node and one or more worker nodes.

G) Perform a stress test to assess the performance of the WEB APP on the cluster by using *wrk*.

H) Deploy the final WEB APP on the Kubernetes cluster just created.

Service Level Agreement

The proposed solution guarantee the following performance

- uptime:project 1: from 6:00 am to 10 p.m. daily, 7 days a week
- project 2, 3: from 8:00 am to 6 p.m. daily, Mon-Fri
- availability rate: >= 99.5 % of the uptime
- outage duration: maximum 30 minutes, average 3 minutes
- reliability rate: >= 99,5% of the uptime
- response time: <= 1 second

Dataset

The dataset to be used for training and testing your solution will be available after the Mid-term break.

Suggested reading

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
Disponibile liberamente in versione HTML autorizzata dagli autori all'indirizzo ufficiale: deeplearningbook.org